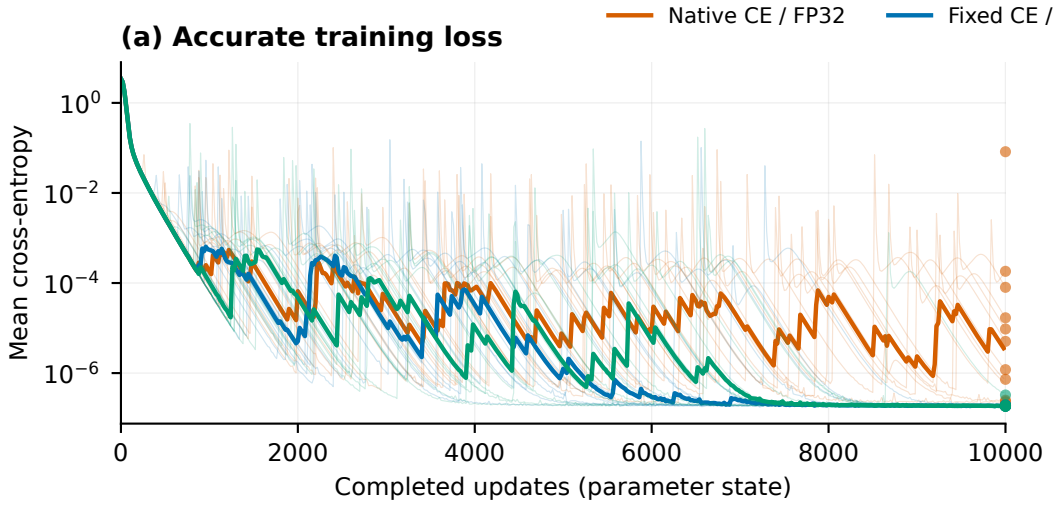
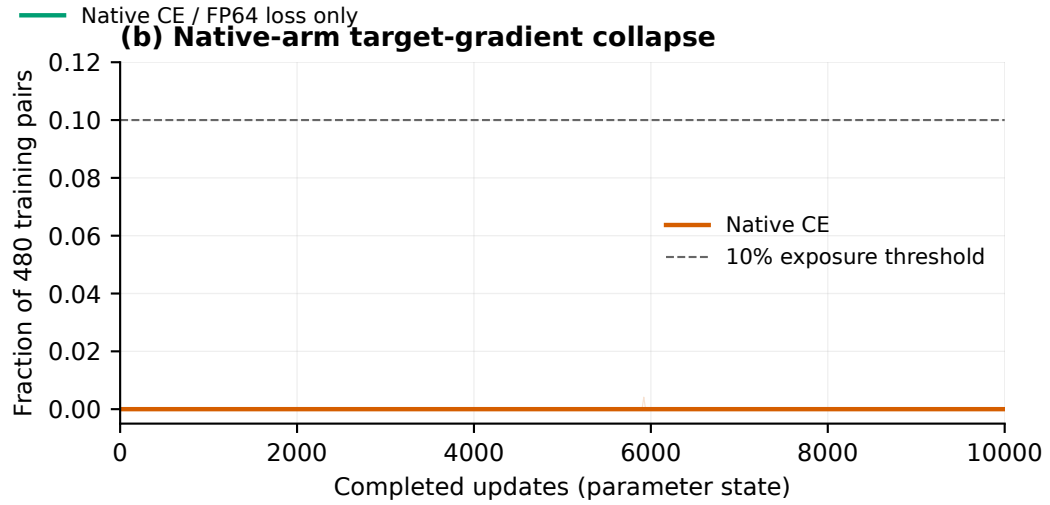
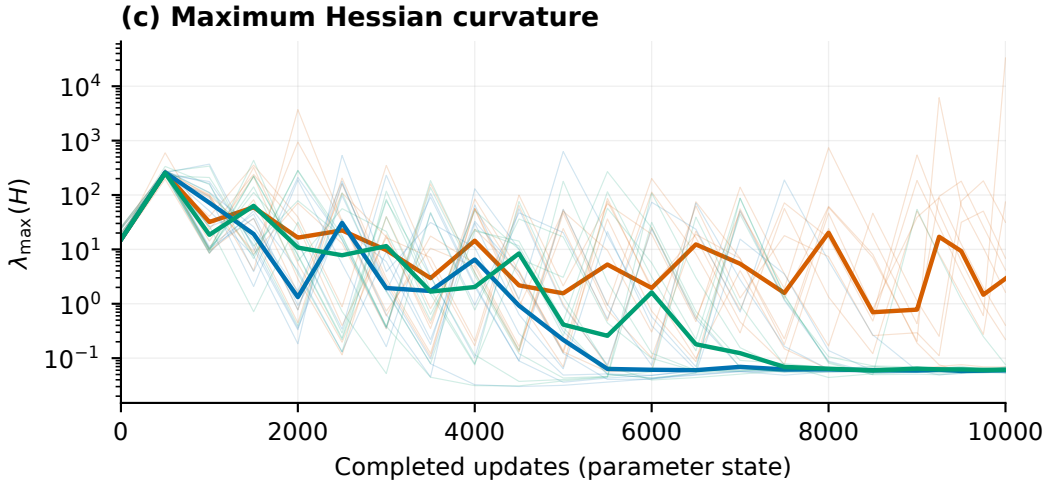
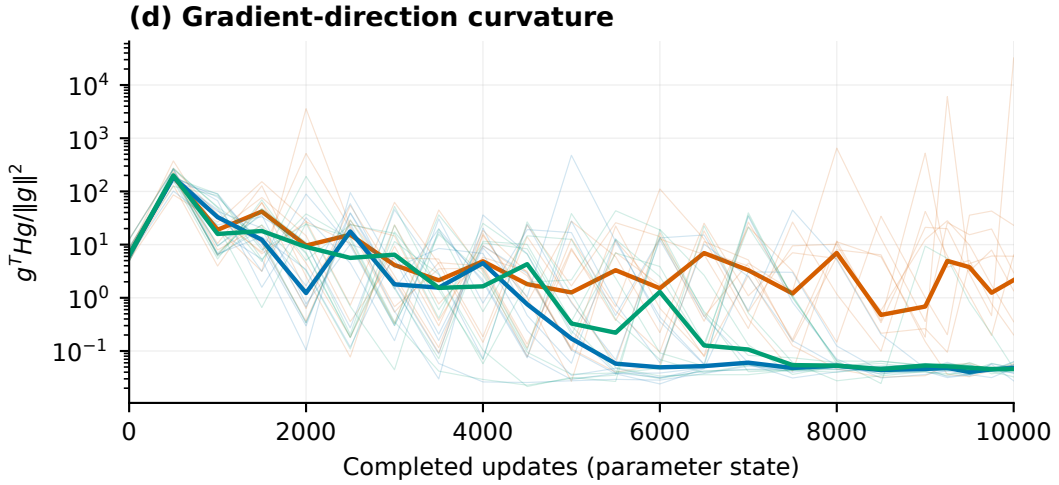
Fixed cross-entropy and curvature under paired AdamW training

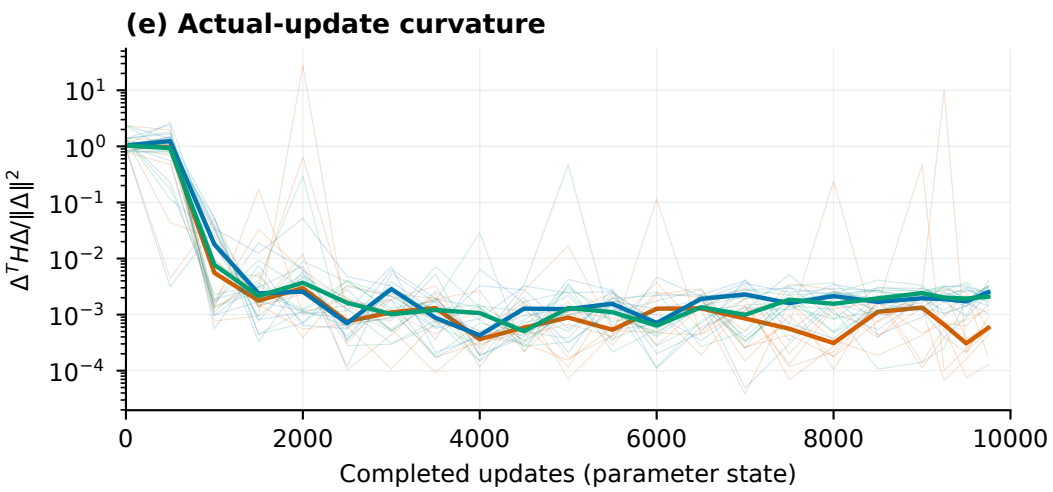
Native CE / FP32 · Fixed CE / FP32 · Native CE / FP64 loss only

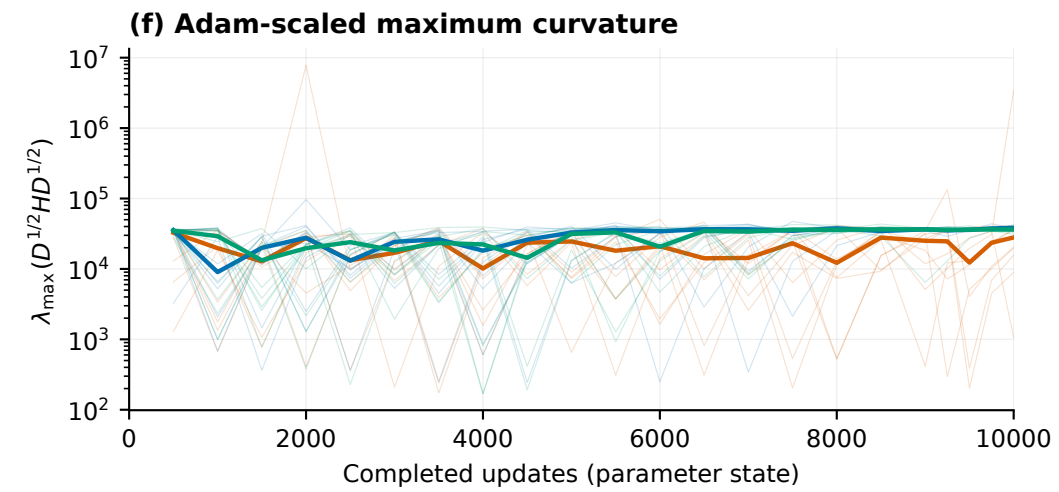


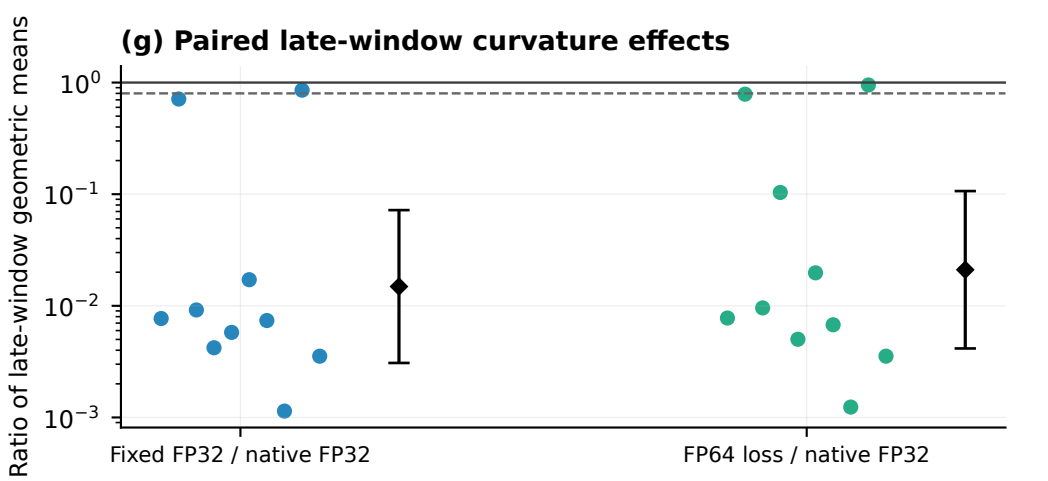


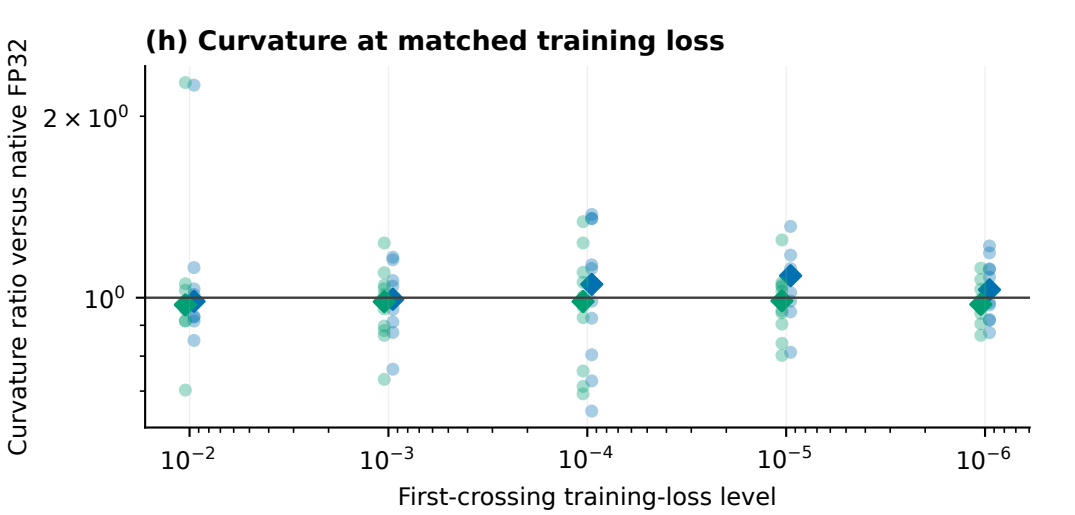












Modular addition mod 31; width-64 Transformer; 10 paired seeds × 3 arms × 10,000 updates. Final dots in (a): full-FP64 objective.
Thin lines: individual seeds; thick lines: medians. Every Hessian uses the same full FP64 objective on all 480 training pairs.
Largest algebraic eigenpairs have relative residual ≤0.005. Collapse exposure qualified in 0/10 native seeds (8 required).
(g) Dots: seeds; diamonds: geometric means + 95% CI; dashed: 20% threshold. (h) Dots: seeds; diamonds: medians; 100/100 loss matches.